

Digital Wallet Payment Integration

eCommerce Platform — India Market

User Stories & Acceptance Criteria

1. Introduction & Project Context

1.1 Project Overview

This document covers the requirements for the digital wallet payment integration on an eCommerce platform serving the Indian market. The platform currently supports card-based payment methods for both guest and registered customers. The wallet payment integration introduces a flexible, account-optional payment channel at checkout, enabling customers to transact using their preferred digital wallet providers.

The integration spans the customer-facing checkout experience, account management for registered users, and operational tooling within the admin panel for the internal Ops team.

1.2 Business Problem

Digital wallet adoption in India has grown significantly, driven by UPI-linked wallets, mobile-first consumer behaviour, and the preference for account-optional payment flows. The platform's current reliance on card-only payments creates unnecessary friction at checkout, particularly for guest users who are unwilling to create an account or enter card details for a single purchase.

This friction contributes to elevated cart abandonment rates and limits the platform's ability to convert first-time visitors into paying customers. The absence of wallet-based payment options also reduces the platform's competitiveness in a market where wallet payments are a standard checkout expectation.

1.3 Scope Summary

The integration covers the following functional areas:

- Guest user wallet payment at checkout — without requiring account creation
- Logged-in customer wallet payment and credential management — including saving and reusing wallet details
- Payment failure handling — with clear customer-facing error messaging and recovery options
- Admin panel configuration — enabling the Ops team to manage wallet payment options, enable or disable providers, and set display preferences
- Audit log and reconciliation tooling — providing the Ops team with visibility into payment failures, exportable records, and automated log lifecycle management

1.4 Document Structure

- Section 2 — Epic statement and business objectives
- Section 3 — Personas involved in the feature
- Section 4 — User stories
- Section 5 — User story to AC traceability table
- Section 6 — Acceptance criteria in Gherkin format
- Section 7 — Assumptions and dependencies

2. Epic Statement & Business Objectives

2.1 Epic

Enable digital wallet payment integration on the eCommerce platform to reduce checkout friction, improve conversion rates, and provide customers with flexible payment options across both guest and logged-in journeys.

2.2 Business Objectives

Business Objective	Rationale
Increase checkout conversion rate	Reducing payment friction for both guest and logged-in users increase the likelihood of order completion
Reduce cart abandonment	Faster, more flexible checkout options lower the rate at which customers exit before completing payment
Improve customer retention	Positive payment experiences and saved wallet convenience encourage repeat purchases

3. Personas

Three distinct personas interact with the wallet payment feature across customer-facing and operational touchpoints.

Persona	Description
Guest User	Non-logged-in customer browsing and purchasing on the platform
Logged-in Customer	Registered customer with a platform account and saved profile
Ops Team Member	Internal operations staff using the admin panel for reconciliation, configuration, and monitoring

4. User Stories

The following user stories were derived from the wallet payment integration epic. Each story is scoped to a single persona and a single user need, following the standard format: As a [persona], I want [action], so that [benefit].

US-01 — Guest User Wallet Payment

As a guest user, I want to select a digital wallet as my payment method at checkout, so that I can complete my purchase without creating an account or entering card details.

US-02 — Logged-in Customer: Wallet Transaction History

As a logged-in customer, I want to see my wallet transaction history in my account, so that I can monitor my wallet transactions effectively.

US-03 — Ops Team: Order List Filtered by Wallet Payment

As an Ops team member, I want to filter and view orders paid via digital wallet from the admin panel, so that I can generate reconciliation reports for wallet-based transactions specifically.

US-04 — Ops Team: Configure Wallet Payment Options

As an Ops team member, I want to configure wallet payment options from the admin panel, so that only the configured wallet payment options are visible on the checkout page.

US-05 — Logged-in Customer: Save Wallet Credentials

As a logged-in customer, I want to save my preferred wallet as a payment method in my account, so that I can complete future purchases faster without re-entering my payment details.

US-06 — Ops Team: Audit Log for Wallet Payment Failures

As an Ops team member, I want to view payment failure logs from the admin panel, so that I can investigate the root cause and take corrective action promptly.

US-07 — Customer: Error Message on Payment Failure

As a customer, I want to see a clear error message when my wallet payment fails at checkout, so that I can understand what went wrong and choose an alternative payment method to complete my order.

5. User Story to AC Traceability

The following table maps each user story to its corresponding acceptance criteria features and scenarios documented in Section 6.

Story ID	User Story Summary	AC Reference
US-01	Guest user wallet payment at checkout	<i>Feature 3 – Scenario 1; cross-ref Feature 2</i>
US-02	Logged-in customer wallet transaction history	<i>No AC defined – straightforward story</i>
US-03	Ops – filter orders by wallet payment	<i>No AC defined – straightforward story</i>
US-04	Ops – configure wallet payment options	<i>Feature 5 – Scenarios 1 & 2</i>
US-05	Logged-in customer – save wallet credentials	<i>Feature 4 – Scenarios 1 & 2</i>
US-06	Ops – view audit log for wallet failures	<i>Feature 4 (Audit Log) – Scenarios 1, 2, 3a & 3b</i>
US-07	Customer – error message on payment failure	<i>Feature 2 – Scenarios 1, 2 & 3</i>

6. Acceptance Criteria

Acceptance criteria are written in Gherkin (Given-When-Then) format. Each feature maps to one or more user stories as referenced in the traceability table above. Scenarios cover happy path, failure path, edge cases, and system-triggered background processes.

6.1 Error Message Display on Wallet Payment Failure

Relates to: US-07

Feature: Error message display at checkout on wallet payment failure

Scenario 1: Wallet authentication failure at checkout

Given a customer has items in their cart
And has proceeded to the payment step at checkout
When they enter their wallet account credentials
And submit the wallet payment details
And the wallet authentication fails
Then an error message is displayed stating the authentication has failed
And the message indicates the customer should verify their wallet credentials
And the checkout page displays available alternative payment methods
And the order is not placed
And the cart contents remain unchanged

Scenario 2: Insufficient wallet balance at checkout

Given a customer has items in their cart
And their cart contains one or more items
When they select wallet as their payment method
And complete wallet authentication successfully
And the payment fails due to insufficient wallet balance
Then an error message is displayed stating the payment failed due to insufficient balance
And the message advises the customer to add funds or select an alternative payment method
And the checkout page displays available alternative payment methods
And the cart contents remain unchanged
And the order is not placed

Scenario 3: Wallet payment API or network failure at checkout

Given a customer has items in their cart
And has proceeded to the payment step
When they select wallet as their payment method
And the system attempts to process the wallet payment
And the payment API fails due to a network or server error
Then an error message is displayed stating the payment could not be processed due to a technical issue
And the message advises the customer to retry or select an alternative payment method
And a retry option is available on the checkout page
And available alternative payment methods are displayed
And the order is not placed
And the cart contents remain unchanged

6.2 Wallet Payment at Checkout for Guest Users

Relates to: US-01

Feature: Wallet payment at checkout for guest users

Scenario 1: Successful order placement using wallet payment

Given a guest user is not logged in to the platform
And has items in their cart
And has proceeded to the payment step at checkout
When they select a digital wallet as their payment method
And enter their wallet account credentials
And submit the payment
Then the payment is processed successfully
And the order is placed successfully
And an order confirmation message is displayed with the order number
And an order confirmation email is sent to the guest user's provided email address

Note: For wallet payment failure scenarios applicable to guest users, refer to
Feature - Error message display at checkout on wallet payment failure.
Failure behaviour is consistent across guest and logged-in users.

6.3 Saving Wallet Credentials for Logged-in Customers

Relates to: US-05

Feature: Saving wallet credentials for logged-in customers

Scenario 1: Successful save of wallet credentials in customer account

Given the customer is logged in to their account
And has navigated to the saved payment methods section
When they select a wallet provider
And enter their wallet account details
And submit the save request
Then the payment provider validates the wallet credentials successfully
And the system displays a confirmation message stating the wallet has been saved
And the saved wallet appears in the customer's saved payment methods
And the saved wallet details show the provider name and masked account information

Scenario 2: Customer reaches maximum failed attempts while saving wallet credentials

Given the customer is logged in to their account
And has navigated to the saved payment methods section
When they select a wallet provider
And enter incorrect wallet account details
And submit the save request
And the customer repeatedly submits with incorrect details
And exhausts the maximum number of save attempts configured by the admin
Then the payment provider validation fails on each attempt
And the system displays an error message stating the maximum save attempts have been reached
And the error message advises the customer to verify their wallet account details
And indicates the duration after which the customer can retry
And the saved payment methods section is locked for the admin configured cooldown period
And the customer retains access to other payment methods and can continue placing orders
And an SMS and email notification is sent to the customer informing them of the lockout
And an audit log entry is created capturing the failed attempts for Ops team review

6.4 Audit Log for Wallet Payment Failures

Relates to: US-06

Feature: Audit log for wallet payment failures in admin panel

Scenario 1: Successful retrieval of failed wallet payment audit logs using filters

Given the Ops team member is logged in to the admin panel with Ops access privileges
When they navigate to the audit log section
And apply filters for a specific date range, payment type as wallet, and status as failed
And submit the filter request
Then the system displays matching log records with pagination
And each log record displays: date and time, payment type, failure status, failure reason,
user type (guest or logged-in), and email address for logged-in users
And for guest users, the log displays the billing email if captured,
or the payment gateway transaction reference ID if email is unavailable
And the Ops team member can navigate through pages to view remaining filtered records

Scenario 2: Audit log pagination and data export for filtered wallet payment failures

Given the Ops team member is logged in to the admin panel with Ops access privileges
When they navigate to the audit log section
And apply filters for a specific date range, payment type as wallet, and status as failed
And select the preferred record count per page from available options of 12, 24, 100, or 200
And submit the filter request
Then the system displays matching log records based on the selected record count per page
And the Ops team member can navigate through pages to view remaining filtered records
And when the Ops team member selects CSV or PDF as the export format
And selects the export scope as current page only or all filtered pages
And the system generates and downloads the export file in the selected format
And the exported file contains all displayed data fields including date and time, payment type, status, failure reason, and user identifier
And the exported file is named with the applied filter criteria and export timestamp

Scenario 3a: Admin configures audit log retention policy and cron schedule

Given the Ops team member is logged in to the admin panel with Ops access privileges
When they navigate to the audit log configuration section
And set the log retention period in days
And configure the cron execution frequency as daily, weekly, or monthly
And set the cron execution timestamp
And save the configuration
Then the system saves the retention policy and cron schedule successfully
And displays a confirmation message stating the configuration has been saved
And the configured cron job is scheduled for execution at the specified frequency and time

Scenario 3b: Scheduled auto-deletion of audit log records based on retention policy

Given the audit log contains records older than the configured retention period
And the retention policy and cron schedule have been configured by the Ops team
And alert notification recipients have been configured in the system
When the scheduled cron job executes at the configured time
Then the system identifies log records that have exceeded the configured retention period
And deletes the identified records from the audit log
And retains records flagged for compliance or regulatory purposes regardless of retention period
And creates a system log entry capturing the number of records deleted and the execution timestamp

And the Ops team member can view only records within the configured retention period

And if the cron job fails, an alert notification is sent to the configured recipients for investigation

6.5 Wallet Payment Configuration in the Admin Panel

Relates to: US-04

Feature: Wallet payment configuration in the admin panel

Scenario 1: Successful wallet payment configuration by Ops team

Given the Ops team member is logged in to the admin panel with Ops access privileges

When they navigate to the wallet configuration section

And complete all required configuration fields including:

- Enable or disable wallet payment
- Available wallet payment options
- Sort order of wallet options
- Wallet payment title
- Message to display when wallet is unavailable
- Enable or disable wallet transaction log
- Default error message for unspecified failures
- Order status on successful wallet payment

And save the configuration

Then the system displays a success message confirming the configuration has been saved

And the enabled wallet payment options are displayed to customers on the checkout payment step

And the wallet options appear in the configured sort order with the configured payment title

And if wallet is disabled, no wallet payment option is displayed to customers at checkout

Scenario 2: Validation failure due to incomplete wallet configuration

Given the Ops team member is logged in to the admin panel with Ops access privileges

When they navigate to the wallet configuration section

And leave one or more required configuration fields incomplete

And attempt to save the configuration

Then the system displays validation errors adjacent to each incomplete required field

And displays a form-level message stating all required fields must be completed before saving

And the configuration is not saved

And the partially entered configuration data is retained on the form for correction

7. Assumptions & Dependencies

7.1 Key Assumptions

- Alert notification recipient configuration is treated as a shared platform capability and is documented as a separate feature. It is referenced by assumption in all scenarios requiring failure notifications, avoiding duplication across stories.
- The payment gateway generates a unique transaction reference ID at the point of payment initiation. This reference ID is available for audit log capture even when no platform order number has been created.
- A compliance flagging mechanism exists within the platform to mark specific audit log records for regulatory retention, preventing their removal during scheduled auto-deletion.
- Maximum save attempt limits, cooldown durations, and log retention periods are all configurable via the admin panel wallet configuration feature (US-04) and are not hardcoded in the system.
- Guest user billing and shipping address capture occurs prior to the payment step. The email address entered in the billing form is therefore available for audit log reference in the event of a guest payment failure.
- Wallet payment failure scenarios for guest users are functionally identical to those for logged-in users. The AC defined under Feature 6.1 applies to both personas. This cross-reference is noted in Feature 6.2 to avoid redundant documentation.

7.2 Out of Scope

- Specific wallet provider integrations (e.g. Paytm, PhonePe, Google Pay) — provider selection is an implementation decision outside the scope of this BRD artifact.
- KYC verification flows for wallet account creation — these are managed by the respective wallet providers.
- COD refund-to-wallet eligibility — deferred to a separate requirements workstream.
- Wallet transaction history acceptance criteria (US-02) and order list filter acceptance criteria (US-03) — these are straightforward stories that do not require detailed AC for portfolio demonstration purposes.

7.3 Dependencies

- Payment gateway API availability and sandbox environment access for integration testing.
- Admin panel role-based access control configuration to support Ops team privilege levels.
- Platform notification service for SMS and email alerts to customers and Ops team members.
- Cron job scheduling infrastructure to support background auto-deletion of audit log records.

This document is a portfolio artifact prepared for BA role applications.